Lab 1 — Command Line

Task 1

Introduce yourself on our Slack channel.

Task 2

Using only a terminal emulator (as described in the Command Line Presentation), carefully perform the following:

1. Create a new directory called `cst205` in your Desktop directory (or a directory of your choosing).
2. From within the `cst205` directory, create a new file called `test.cst205`
3. On Windows, open this file from PowerShell using Notepad, with: `notepad test.cst205`

   (Alternatively, on **Windows**, (if you feel comfortable doing so) you can install Vim. Do the "full install" and then you can access via PowerShell.)

   On **macOS/Linux**, open this file using the command line editors `nano` or `vim`. (Ask for help if needed!)
4. Type a few fun facts about the other students in the class and save and close the file.
5. Rename the file as `hello.205` (using the command line)
6. Verify the file contents using the `cat` command
7. Issue the following command(s) based on the operating system you are using. For easier legibility, refer to the this GitHub Gist.

   On Windows, issue the following command:

   ```
   0..2 | % { Get-Random -min 0 -max 255} > color.txt
   ```

On macOS / Linux, issue the following commands:

```
for ((n=0; n<3; n++));
do echo $((RANDOM % 255));
done > color.txt
```

8. `cat` the file and enter the three values as the R, G, B values on this website. On the fourth line of color.txt, write down the hex value provided (starts with #), for example, `#84D0D7`.

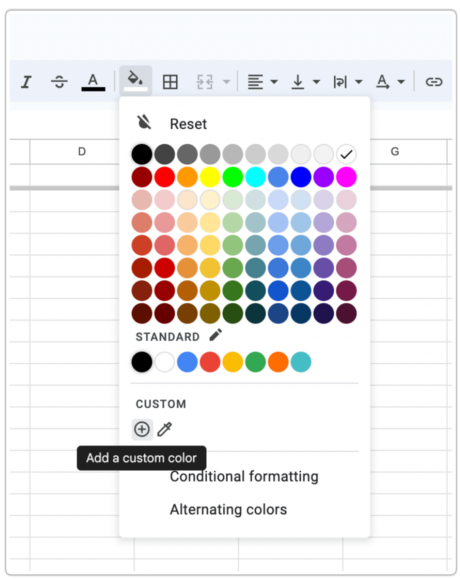
9. Enter in your hex value (without the #) at the end of the URL: `https://www.colorhexa.com/` to learn more about your color. For example, if your color is `#84D0D7`, you will go to the following URL:

   `https://www.colorhexa.com/84D0D7`

10. Go to the the following spreadsheet and find the row with your email address. In the column next to it, enter your preferred first name AND add your color as follows:

    With the cell selected, choose Fill Color and then "Add custom color"

    In the dialog you will be able to enter your color either as RGB or Hex. This will change the background of the cell to that color. Feel free to change the text color with your name to something that makes it legible. Row 1 has been done as an example.



Task 3

For your respective operating system, research and explain how the commands from Task 2 work. You can add this information to your `color.txt` file.

Task 4

For this task, we want to use the command line to explore if Python and Git are installed.

On **macOS/Linux** use the `which` command as follows: `which python`
If this returns ("python not found") try which `python3`

On **Windows**, use `Get-Command` as follows: `(Get-Command python).Path`
If this gives you an error, try `(Get-Command py).Path`

Do the same for `git` (no variations needed).

Deliverable

Submit `hello.205`, `color.txt`, and a screenshot of your terminal emulator (at the end of the lab) on Canvas.